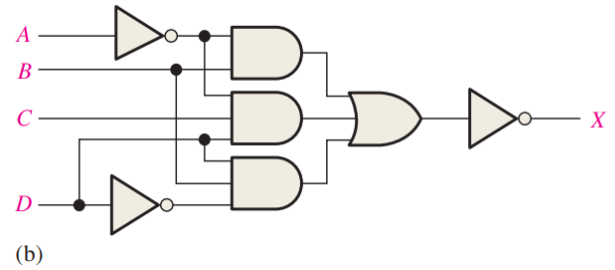
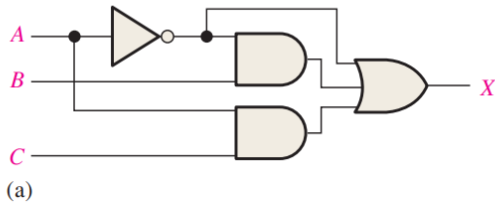


Chapter 5

2024.2.22

Name:Chen Zhuoyan Student ID:523030910037

2. Write the output expression for each circuit in the Figure.



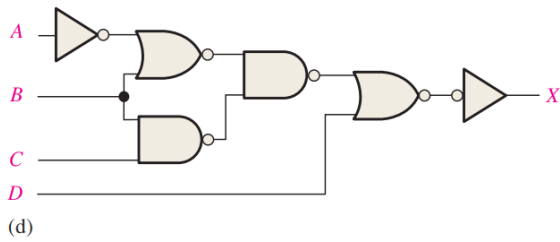
Answer Area:

(a) $X = \overline{A} + \overline{A}B + AC$

(b) $X = \overline{\overline{A}B + \overline{A}CD + BDD}$

6. Develop truth table for each circuit in the Figure:

(d)



Answer Area:

(d)

$$X = (\overline{A} + B)\overline{BC} + D$$

A	B	C	D	X
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

7. Show that an exclusive-NOR circuit produces a POS output.

Answer Area:

∴ The truth table of exclusion-NOR gate is as below:

A	B	X
0	0	1
1	0	0
0	1	0
1	1	1

∴ $A \odot B$ is equivalent with $(\overline{A} + B)(A + \overline{B})$

∴ An exclusive-NOR circuit produces a POS output.

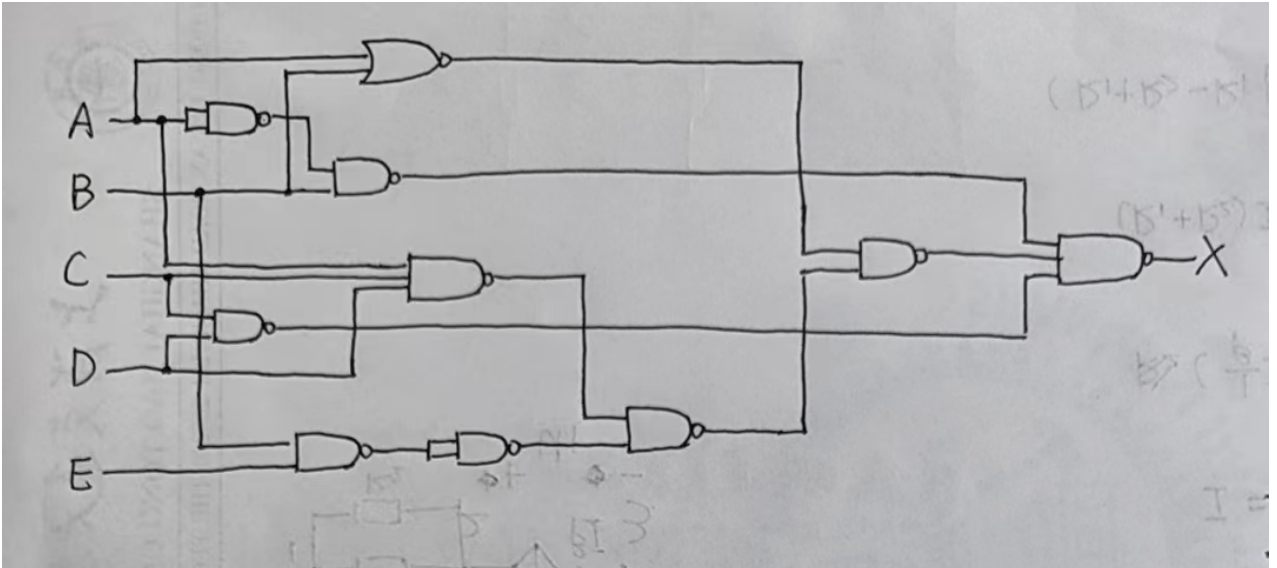
12. Use NAND gates, NOR gates, or combinations of both to implement the following logic expressions as stated :

(a) $X = \overline{A}B + CD + (\overline{A + B})(ACD + \overline{BE})$

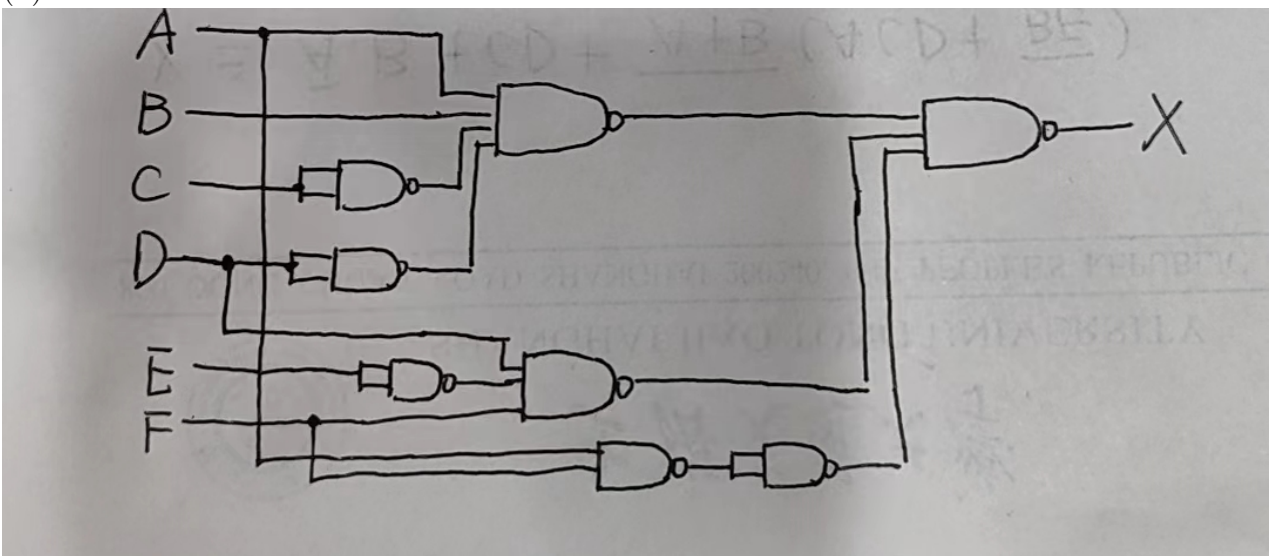
(d) $X = ABC\overline{C} \cdot \overline{D} + D\overline{E}F + \overline{A}F$

Answer Area:

(a)



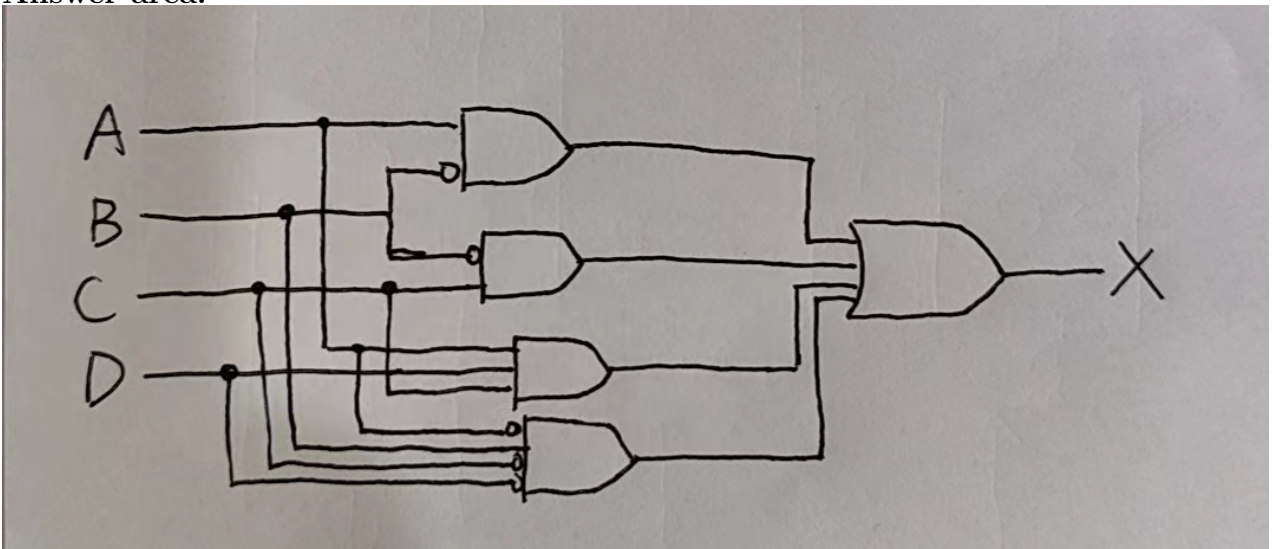
(d)



14. Implement a logic circuit for the truth table in Table 5-9:

TABLE 5-9				
Inputs				Output
A	B	C	D	X
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Answer area:



21. Implement the logic circuit in Figure 5-58 using only NAND gates.:

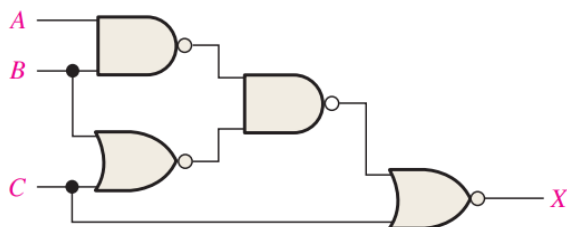
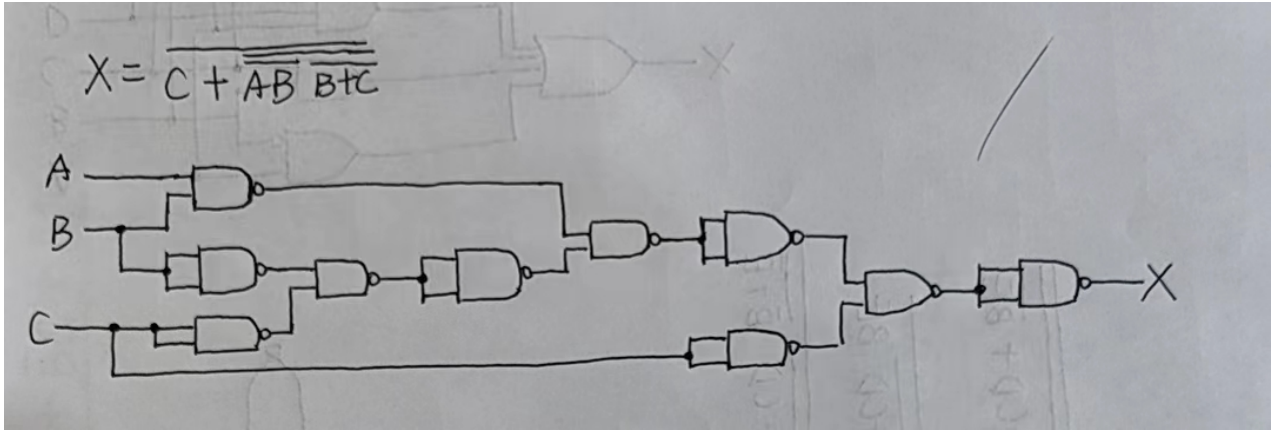


FIGURE 5-58

Answer Area:



24. Show how the following expressions can be implemented as stated using only NOR gates:

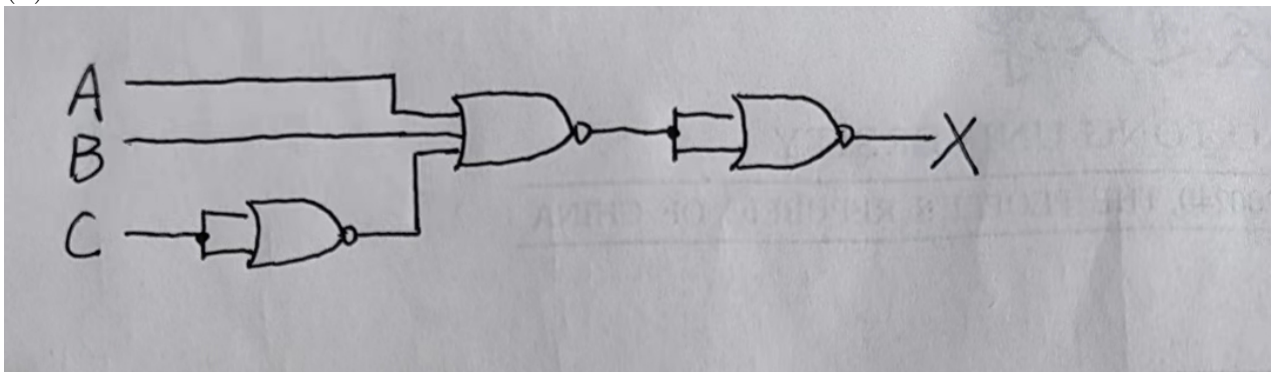
(d) $X = A + B + \overline{C}$

(f) $X = (A + B)(C + D)$

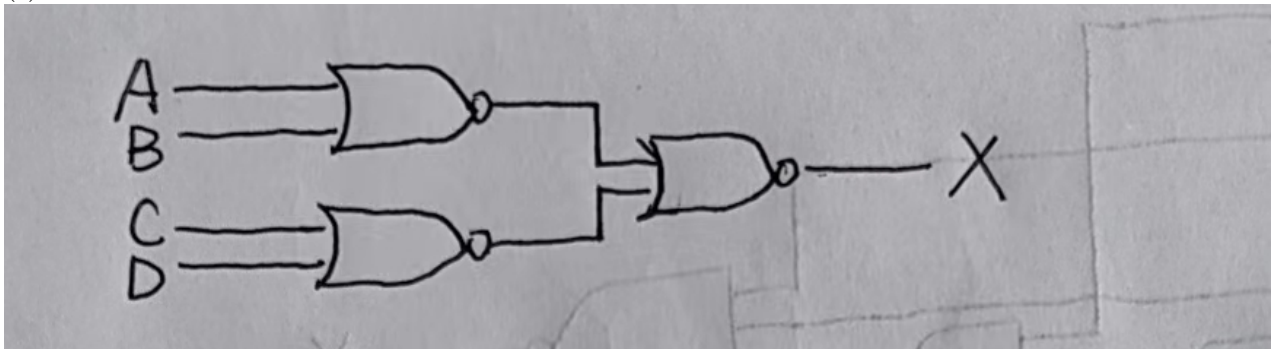
(g) $AB[C(\overline{DE} + \overline{AB}) + \overline{BCE}]$

Answer Area:

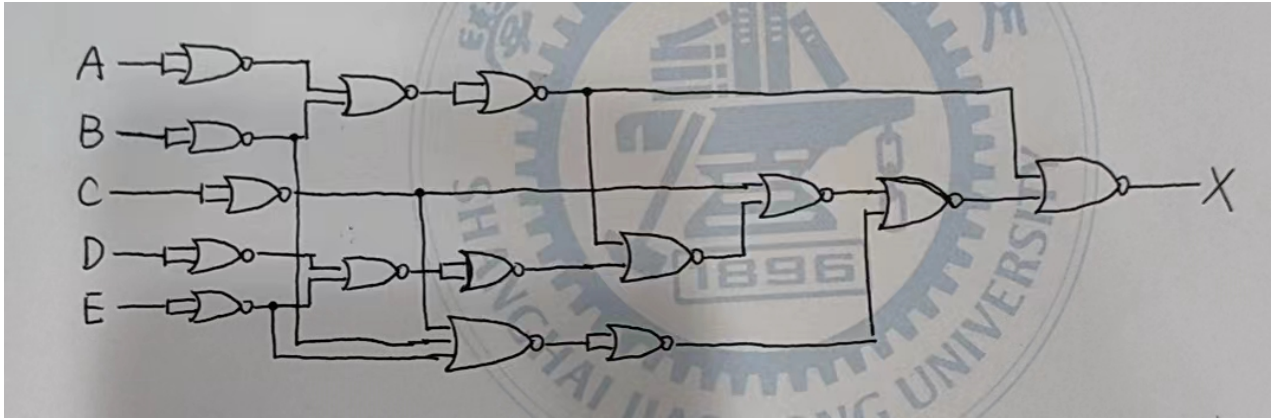
(d)



(f)



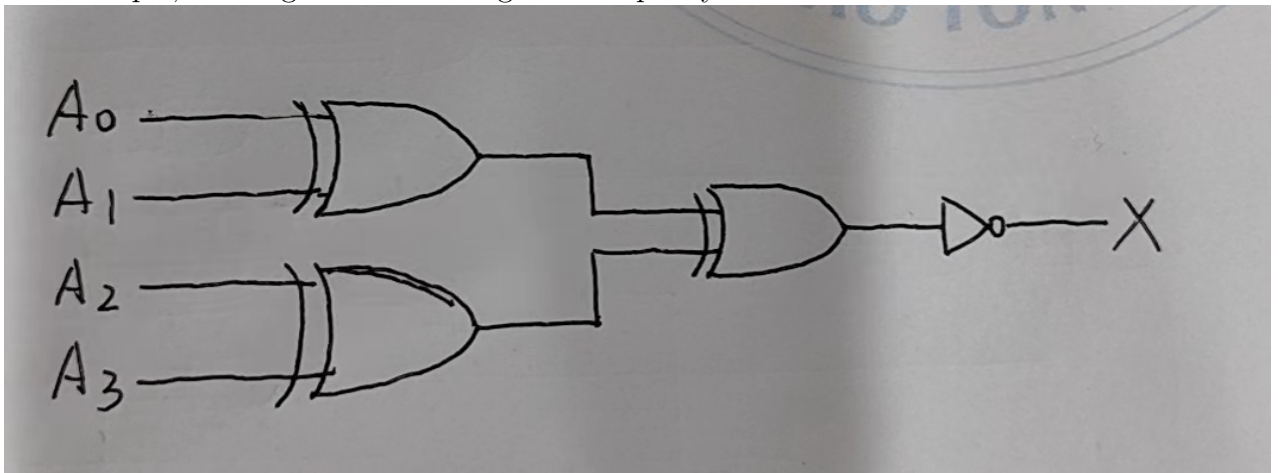
(g)



Part 2:

How to design a even-parity checker?

For example, to design a four-bits signal even-parity checker:



To design such checker, you need to pair the signals and let them pass exclusive-OR gates. Repeat the step until there is only one output. Then let the output pass a NOT gate.